Industrial Internship Report on

"Prediction of Agriculture Crop Production in India"

Prepared by:

Anjana Prakash

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project: 1.Prediction of Agriculture Crop Production in India This project focuses on This project focuses on predicting agricultural crop production in India using historical data from 2001–2014. The dataset includes crop type, variety, state, season, production quantity, cultivation cost, production, and recommended cultivation zones. Machine learning techniques are used to analyze the relationship between these factors and accurately forecast crop production. The prediction model assists farmers, agricultural planners, and government agencies in making informed decisions regarding crop selection, resource allocation, cultivation planning, and food security. It also helps identify high-yield crops, estimate production levels, and improve agricultural productivity while minimizing risks associated with changing environmental and economic conditions. The project supports data-driven decision-making for sustainable agricultural development in India. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

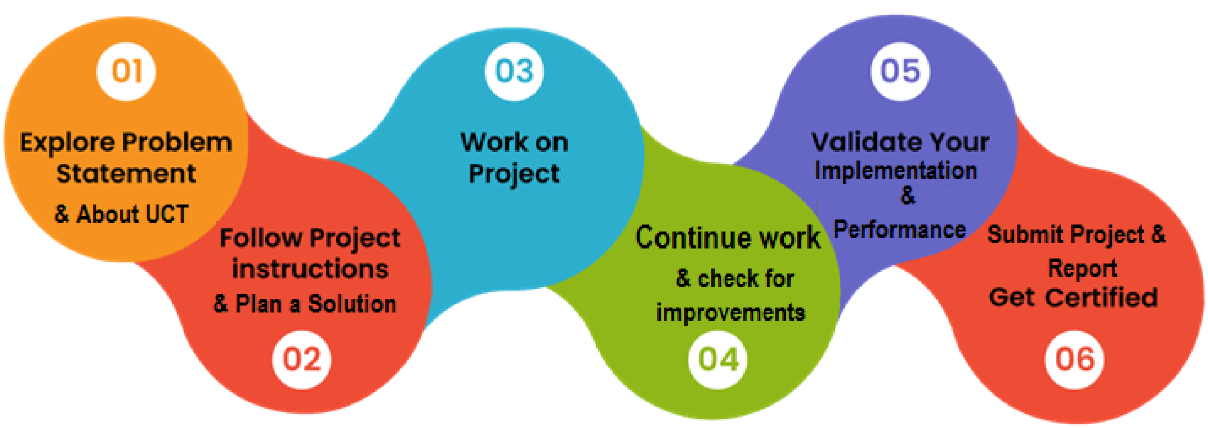
1 Preface

The six-week internship was a valuable learning experience that allowed me to apply data science and machine learning concepts to real-world problems. During the internship, I learned the complete machine learning workflow, including data preprocessing, exploratory data analysis (EDA), model building, evaluation, and result interpretation using Python. I worked on project: **Prediction of Agriculture Crop Production in India**, which aimed to forecast crop production using historical agricultural data to assist farmers and policymakers in decision-making. The internship also helped me improve my analytical thinking, problem-solving, technical documentation, and time management skills. Overall, it strengthened my practical knowledge of machine learning and increased my confidence in developing data-driven solutions for real-world applications.

A relevant internship plays an important role in career development by providing practical exposure to real-world problems and industry practices. It helps bridge the gap between classroom learning and professional work, allowing students to apply their technical knowledge, develop problem-solving abilities, and improve communication and teamwork skills. It also builds confidence and prepares students for future job opportunities.

The internship offered by USC/UCT provided an excellent platform to gain hands-on experience in data science and machine learning. It gave me the opportunity to work on real-world datasets, learn industry-relevant tools and techniques, and understand the complete project development process. The guidance provided throughout the internship helped me strengthen both my technical knowledge and professional skills, making the experience valuable for my academic and career growth.

How Program was planned:



The internship enhanced my understanding of data science and machine learning through practical implementation. I learned how to preprocess data, perform exploratory data analysis, build and evaluate machine learning models, and interpret results effectively. Working on real-world projects improved my analytical thinking, problem-solving abilities, and confidence in applying theoretical concepts to practical situations. Overall, the internship was a rewarding experience that contributed significantly to my technical and professional growth.

I would like to express my sincere gratitude to USC/UCT for providing me with this valuable internship opportunity. I am thankful to my project mentors and trainers for their constant guidance, encouragement, and support throughout the internship. I also extend my heartfelt thanks to the faculty members of my college, my classmates, friends, and my family for their motivation and support, which helped me successfully complete this internship.

I encourage my juniors and peers to make the most of internship opportunities, as they provide valuable practical experience beyond classroom learning. Be curious, ask questions, stay consistent, and never hesitate to explore new technologies. Every project, no matter how small, is an opportunity to learn and grow. Continuous learning and hands-on practice are the keys to building a successful career in technology.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

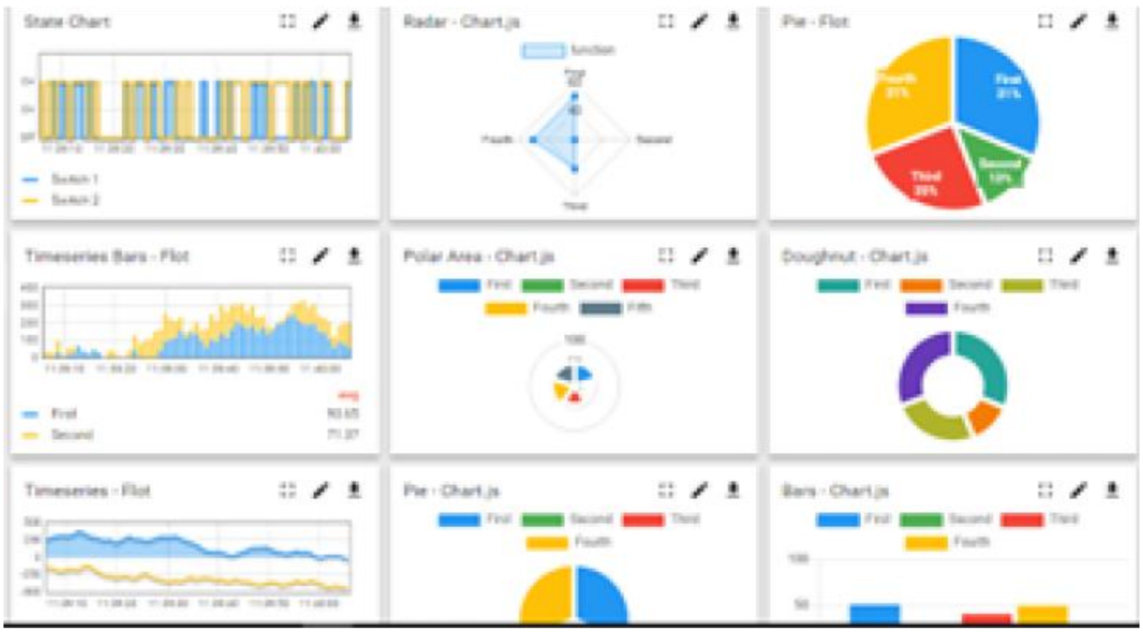
i. UCT IoT Platform (Insight)

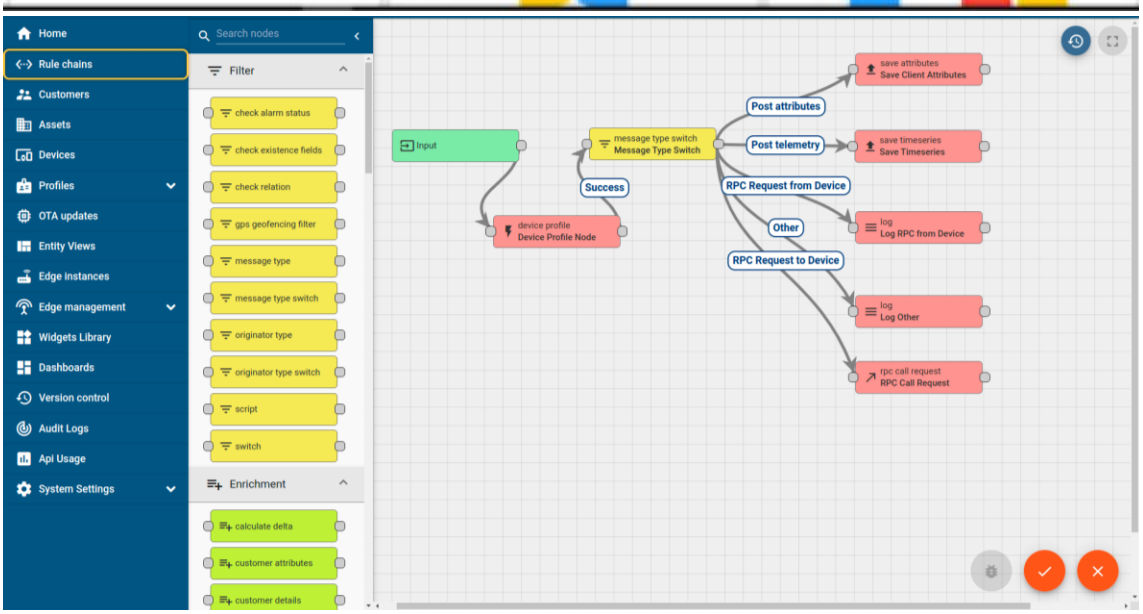
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. Smart Factory Platform ( )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

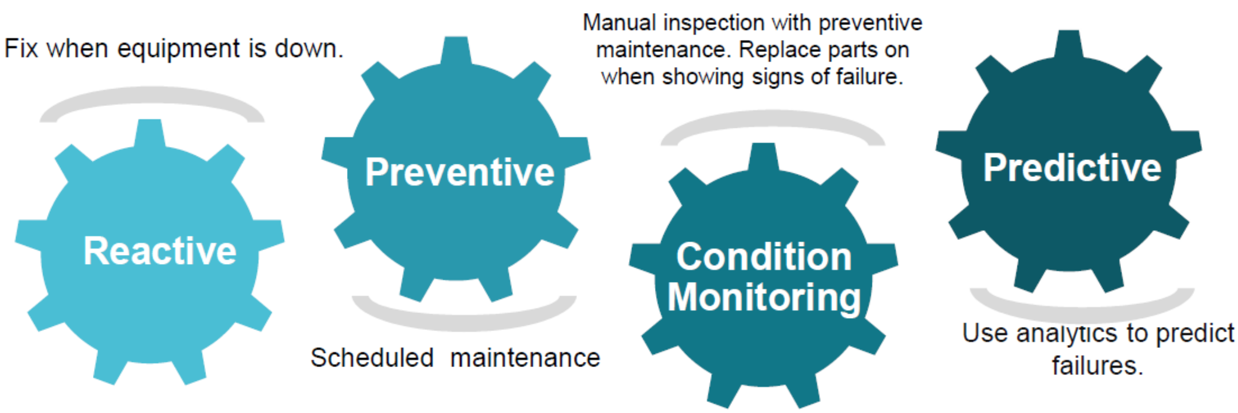
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

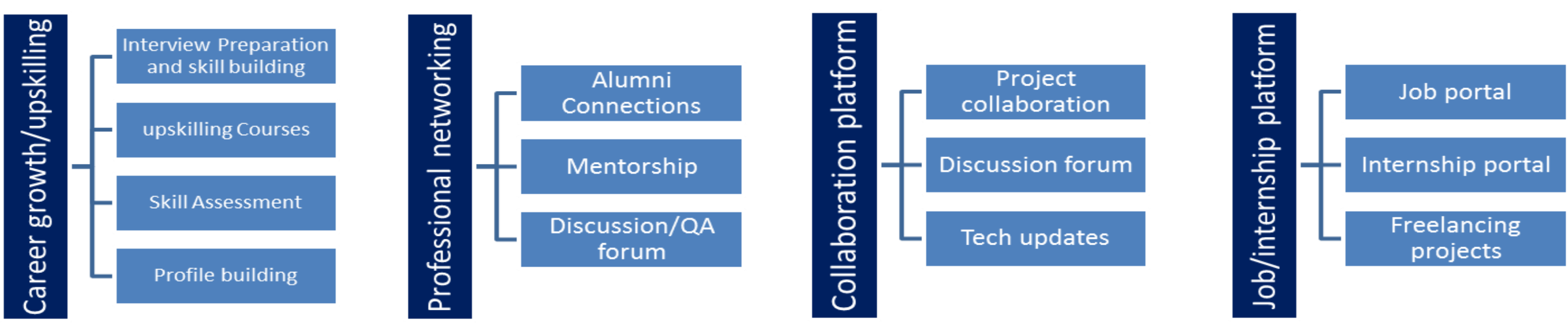
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

3 Problem Statement

During the internship, I worked on two real-world machine learning problem statements.

The problem statement, **Prediction of Agriculture Crop Production in India**, focused on predicting crop production using historical agricultural data. The dataset contained information such as crop type, variety, state, season, production, cultivation cost, and recommended cultivation zones. The objective was to build a predictive model that estimates crop production accurately, enabling farmers and policymakers to make informed decisions for improving agricultural productivity and resource management.

Problem statements demonstrated how machine learning can be applied to solve practical challenges and support data-driven decision-making in the fields of urban development and agriculture.

4 Existing and Proposed solution

4.1 Existing Solutions

Existing crop production prediction systems mainly depend on historical production records, statistical methods, and manual analysis by agricultural experts. Farmers also rely on their experience and traditional farming practices to estimate crop yield. While these methods provide useful insights, they often do not consider multiple influencing factors simultaneously, such as seasonal variations, cultivation cost, regional differences, and changing agricultural conditions. As a result, the predictions may not always be accurate or timely.

4.2 Proposed Solution

The proposed solution uses machine learning techniques to predict agricultural crop production by analyzing historical data. The model is trained using features such as crop type, variety, state, season, cultivation cost, and historical production records to identify patterns that influence crop yield. After data preprocessing and feature selection, the trained model generates accurate production forecasts, enabling farmers and government agencies to make informed decisions regarding crop planning, resource allocation, and agricultural management

4.3 Value Addition

The proposed system provides more accurate and faster crop production predictions compared to traditional methods. It helps farmers choose suitable crops for a particular season and region, supports policymakers in agricultural planning, and assists in efficient resource utilization. The system can reduce uncertainty in crop planning, improve productivity, and contribute to better food security.

4.4 Code submission (Github link)

https://github.com/anjanaprakash2906/upskillcampus/blob/main/PredictionOfAgricultureCropProductionInIndia.py

4.5 Report submission (Github link) : first make placeholder, copy the link.

https://github.com/anjanaprakash2906/upskillcampus/blob/main/PredictionOfAgricultureCropProductionInIndia_Anjana%20Prakash_USC_UCT.pdf

5 Proposed Design/ Model

The proposed solution follows a standard machine learning workflow to predict agricultural crop production. The process begins with collecting the historical agricultural dataset, followed by data preprocessing to handle missing values, remove inconsistencies, and prepare the data for analysis. Exploratory Data Analysis (EDA) is performed to identify trends and relationships among features such as crop type, state, season, cultivation cost, and production. Relevant features are selected, and the dataset is divided into training and testing sets. A machine learning algorithm is then trained, evaluated using appropriate performance metrics, and used to predict future crop production. The predicted results can support farmers and policymakers in making informed agricultural decisions.

5.1 Interfaces (if applicable)

The system accepts historical agricultural datasets in CSV format as input. The data is processed using Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. The workflow includes data preprocessing, feature engineering, model training, prediction, and result visualization. The output consists of predicted crop production values along with charts and graphs that help users understand production trends and make better farming and planning decisions.

6 Performance Test

The performance of the proposed **Agriculture Crop Production Prediction** model was evaluated to determine its effectiveness in predicting crop production using historical agricultural data. The key constraints identified were **prediction accuracy, data quality, computational time, and scalability**.

To improve model performance, the dataset was preprocessed by handling missing values, removing inconsistencies, and selecting relevant features such as crop type, state, season, cultivation cost, and historical production. The trained model was evaluated using regression metrics such as **Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score**. The results indicated that the model was able to predict crop production with satisfactory accuracy.

Although the model performed well on the available dataset, prediction accuracy may be affected by incomplete or outdated data. Future improvements can be made by integrating real-time weather information, soil characteristics, rainfall data, and advanced machine learning algorithms to improve accuracy and practical usability.

6.1 Test Plan/ Test Cases

The machine learning models were tested using historical datasets after data preprocessing and feature selection. The dataset was divided into training and testing sets to evaluate the model's prediction capability.

Test Case	Objective	Expected Result
Data Validation	Verify dataset quality after preprocessing	Clean and consistent dataset
Model Training	Train the model using training data	Successful model generation
Prediction Test	Predict crop production using test data	Accurate predictions
Performance Evaluation	Measure model performance using evaluation metrics	Low prediction error and good accuracy
Result Verification	Compare predicted values with actual values	Predictions closely match actual data

6.2 Test Procedure

1. Collect and preprocess the dataset by removing missing values and inconsistencies.
2. Perform Exploratory Data Analysis (EDA) and feature selection.
3. Split the dataset into training and testing sets.
4. Train the machine learning model using the training data.
5. Test the model using unseen test data.
6. Evaluate the model using performance metrics such as MAE, RMSE, and R^2 Score.
7. Analyze the prediction results and compare them with actual values.

6.3 Performance Outcome

The developed models produced reliable predictions for both traffic forecasting and crop production. Data preprocessing and feature engineering improved the prediction accuracy, while regression evaluation metrics indicated satisfactory model performance. The models successfully identified trends in historical data and demonstrated their usefulness for real-world decision-making. Future improvements, such as integrating real-time data and advanced algorithms, can further enhance prediction accuracy and system performance.

7 My learnings

The internship was a valuable learning experience that helped me strengthen both my technical and professional skills. I gained hands-on experience in data preprocessing, exploratory data analysis (EDA), machine learning model development, performance evaluation, and result interpretation using Python. Working on real-world projects improved my analytical thinking, problem-solving ability, and understanding of the complete machine learning development lifecycle.

This internship also enhanced my confidence in applying theoretical concepts to practical applications and taught me the importance of teamwork, time management, and technical documentation. The knowledge and experience gained during this internship have prepared me for future academic projects, internships, and career opportunities in the fields of Artificial Intelligence, Machine Learning, and Data Science. Overall, the internship has provided a strong foundation for my professional growth and motivated me to continue learning and developing industry-relevant skills.

8 Future work scope

Although the proposed model achieved satisfactory results, there is significant scope for future enhancement. Due to the limited duration of the internship, several advanced features could not be implemented.

Future improvements include integrating **real-time weather data, soil quality, rainfall, humidity, fertilizer usage, and satellite imagery** to improve the accuracy of crop production prediction. The system can also be extended to provide **crop recommendation**, **fertilizer recommendation**, and **market price prediction**, helping farmers make better cultivation and marketing decisions.

Advanced machine learning and deep learning algorithms, such as **XGBoost, Random Forest, LSTM, and ensemble models**, can be explored to further improve prediction performance. Additionally, developing a **web or mobile application** would make the system more accessible to farmers and agricultural officers.

With these enhancements, the proposed solution can evolve into a comprehensive decision-support system that promotes **smart farming, efficient resource utilization, increased agricultural productivity, and sustainable agricultural development**.